



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

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SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

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1. CHEMICAL IDENTIFICATION

Product Identification

Trade name: Bactospeine FC

Product name: *Bacillus thuringiensis* subsp. *Kurstaki* 6.58 x 10⁹ cfu/ml (10,600 IU/mg) SC

Product use: Insecticide

Relevant identified uses of the substance/mixture and uses advised against:

Used as agricultural insecticide. It is a violation of Federal law to use this product in a manner inconsistent with its FIFRA pesticide labeling.

2. HAZARD IDENTIFICATION

Classification of the Substance or Mixture

Eye Irritation – Category 2B

Skin Sensitization – Category 1B

Labeling Elements Symbol(s)



Signal Word: WARNING

Hazard Statement(s): Causes eye irritation

May cause an allergic skin reaction

Precautionary Statement:

Prevention

Avoid breathing mist/spray/vapors

Wash hands and face thoroughly after handling

Wear protective clothing/eye protection

Contaminated work clothing must not be allowed out of the work place

Wear protective gloves.



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Response

IF IN EYES:	Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
If eye irritation persists:	Get medical advice/ attention.
If on skin:	Wash with plenty of water
If skin irritation or rash occurs:	Get medical advice/attention. Take off contaminated clothing and wash it before reuse. Wash contaminated clothing before reuse.

Disposal

Dispose of contents/container in accordance with local/regional/national regulations.

Safety

Avoid contact with skin and eyes.

In case of contact with skin or eyes, rinse immediately with plenty of water.

Wear suitable protective clothing and eye/face protection.

Adverse Human Health Effects

Exposure may result in eye irritation. A skin sensitization (allergic) reaction may occur in sensitive individuals.

Note to Physician:

There is no specific antidote if this product is ingested. If a large amount has been ingested and emesis is inadequate, lavage stomach. Treat symptomatically.

3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical Name: *Bacillus thuringiensis* subsp. *kurstaki*

EPA Regulation Number: 73049-46

List Number: 60178

Chemical Family: Microbial, Btk strain ABTS-351

Ingredients and Composition:

Component	CAS No.	%
Active ingredient		
- <i>Bacillus thuringiensis</i> subsp. <i>kurstaki</i> Strain ABTS-351 fermentation solids	68038-71-1	17.19
Other ingredients	-	82.81



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4. FIRST AID MEASURES

Description of first aid measures

General:

In all cases of doubt, seek medical attention.

In case of swallowed:

Remove from source of exposure. Move person to fresh air. Do NOT induce vomiting. Give large quantities of water. If signs of toxicity occur, seek medical attention. Provide symptomatic or supportive care as necessary.

In case of eyes contact:

Remove from source of exposure. Flush with copious amounts of water. Remove contact lenses, if present and easy to do, after the first 5 minutes, then continue rinsing. If irritation persists or signs of toxicity occur, seek medical attention. Provide symptomatic or supportive care as necessary.

In case of skin contact:

Remove from source of exposure. Take off contaminated clothing. Flush with copious amounts of water. Cover irritated skin with an emollient. If irritation persists or signs of toxicity occur, seek medical attention. Provide symptomatic/supportive care as necessary.

In case of Inhaled contact:

Remove from source of exposure. If signs of toxicity occur, seek medical attention. Provide symptomatic/supportive care as necessary. If person is not breathing, call 911, then give artificial respiration.

Most important Symptoms and Effects, both Acute and Delayed Acute

Acute: Eye irritation and skin sensitization

Delayed acute: No information on significant adverse effects

Indication of any Immediate Medical Attention and Special Treatment Needed

Treat symptomatically and supportively

5. FIRE FIGHTING MEASURES

Lower Flammability limits (LFL): Not Determined

Upper Flammability limits (UFL): Not Determined

Extinguishing Media:

Dry chemical, water spray, foam or carbon dioxide. Use appropriate medium for the underlying cause of the fire.

Specific Hazards Arising from the Chemical:



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None known

Thermal decomposition products:

Carbon oxides, nitrogen oxides

Fire and Explosion hazards:

Not expected to be flammable.

Advice to Firefighters Protective Equipment and precautions for firefighters:

Fire fighter should wear full-face, self-contained breathing apparatus and impervious protective clothing. Fire fighters should avoid inhaling combustion products. See Section 8 (Exposure Controls or Personal Protection)

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Personal Precautions, Protective Equipment and Emergency Procedures

Use in a well-ventilated area. Respiratory protection such as a dust mask should be worn during normal product use. Wear eye protection appropriate to handling activities. Wear impervious gloves. Wear appropriate body coverings if contact may occur. Where exposure level is not known, wear respirator protection suitable for expected level of exposure, in addition to the personal protection as recommended in Section 8.

Environmental Precautions

Keep out of drains, sewers, ditches and waterways. Avoid runoff to waterways and sewers. Dispose of excess product and wash waters according to local regulations.

Methods and Materials for Containment and Cleaning Up

Ventilate area. Wear appropriate personal protective equipment. Recover product with inert material (earth, sand, vermiculite), place into appropriate container for disposal. Do not flush to sewer. Avoid generating dust.

Reference to Other Sections

See Section 7 for Handling Procedures. See Section 8 for Personal Protective Equipment recommendations. See Section 13 for Disposal Considerations.

US Regulations may require reporting spills of hazardous materials: See Section 15: REGULATORY INFORMATION for details on reportable quantities, if any.

7. HANDLING AND STORAGE

Good industrial practice in housekeeping and personal hygiene should be followed.



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Precautions for Safe Handling:

Ventilate. Avoid breathing dust or vapor. Avoid contact with eyes, skin or clothing. Wash thoroughly after handling. Do not eat, drink or smoke while working with product, obey reasonable safety precautions and practice good housekeeping. For filling operations, respiratory protection may be recommended, particularly in enclosed areas.

Storage Conditions:

Protect against physical damage. Close containers of unused material. Store in a dry, cool, ventilated place, away from direct sunlight. Store in a cool, dry place. Keep containers tightly closed when not in use. Store at temperatures above freezing and below 32 °C (90 °F).

Incompatibilities:

None Known

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Exposure standards

Control Parameters Component Exposure Limits

None

Derived No Effect Levels (DNELs)

No DNELs available

Predicted No Effect Concentrations (PNECs)

No PNECs available

Engineering controls

Provide general ventilation. Where adequate general ventilation is unavailable, use process enclosure, local exhaust ventilation, or other engineering controls.

SKIN PROTECTION:

Wear protective clothing, including boots and gloves. Wear gloves made of Latex or other impervious materials. Wash thoroughly with soap and water after handling.

EYE PROTECTION:

Wear goggles, safety glasses with side shields or full-face shield when splashing or spraying of materials is likely.

RESPIRATORY PROTECTION:

For filling operations, if dust/mist is produced, respiratory protection is recommended. Where respiratory protection is warranted, use a dust/mist filtering respirator (MSHA/NIOSH approved, number prefix TC-21C or a NIOSH approved respirator with any N, P, R or HE filter). Wash thoroughly with soap and water after handling.



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9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance:	Liquid, opaque, light brown	Odor:	Musty
pH:	≈4.5- 4.6 (1% dilution)	Odor Threshold:	Not Determined
Melting Point:	Not Applicable	Boiling Point:	Not Determined
Specific Gravity:	Not Determined	Solubility (H ₂ O):	Not Determined
Octanol/H ₂ O Coeff:	Not Determined	Relative Density:	1.1 ± 0.1 g/ml
Molecular Weight:	Not Applicable	Evaporation Rate:	Not Applicable
Auto Ignition:	Not Determined	Decomposition Temp.:	Not Determined
Flash Point:	Non-flammable	LFL:	Not Determined
Vapor Density:	Not Determined	UFL:	Not Determined
VOC:	Aqueous	Vapor Pressure:	Not Determined
Kinematic Viscosity:	Not Determined	Flammability Class:	Not flammable

10. STABILITY AND REACTIVITY

Reactivity:

Material does not pose a significant reactivity hazard.

Chemical Stability:

Stable under ordinary conditions of use and storage. Spontaneous reaction not possible.

Possibility of Hazardous Reactions:

Does not undergo hazardous polymerization

Conditions to Avoid:

None known

Incompatible Materials:

None known

Hazardous Decomposition Products:

Carbon oxides and unidentified organic compounds.

11. TOXICOLOGICAL INFORMATION

Likely Routes of Exposure

Inhalation: Yes

Eye Contact: Yes

Skin Contact: Yes

Ingestion: Not likely



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Product (or TGAI where noted) Toxicology Acute Effects

Oral Toxicity:	LD50	> 5,000 mg/kg (rat) [EPA Toxicity Category IV]
Dermal Toxicity:	LD50	> 5,000 mg/kg (rat)
Inhalation:	LC50 (4h)	> 2.14 mg/L (rat)
Skin irritation:		Mildly irritating, [EPA Toxicity Category IV]
Eye irritation:		Mildly irritating, (cleared by day-7)]
Skin Sensitization:		Sensitizer (guinea pig); undiluted material
Mutagenicity Information:		Components of this product are not listed as mutagens.
Carcinogenicity Information:		Components of this product are not listed as carcinogenic by NTP, IARC or OSHA.
Developmental/Reproductive Toxicity:		This material is not teratogenic
Special Target Organ:		Eye; Skin
Aspiration Hazard:		Not Applicable
Repeat Dose Studies:		Not Applicable, Acute Toxicity testing was all negative

12. ECOLOGICAL INFORMATION

Ecotoxicity

Data referred to the mixture, Bactospeine FC:

Fish – Acute toxicity:

Highly toxic to fish and aquatic invertebrates

Rainbow trout,

LC50 (96 h): >2.87 x 10⁷ cfu/L

Daphnia magna (a small freshwater crustacean),

EC50 (48 h): >50 mg/L

NOEC (21-day) >5.0 mg/L

Birds – Acute toxicity:

Bobwhite quail,

LC50 (5 days): >2,857 mg/kg (5.7 x 10¹⁰ cfu/kg)

Honey bee – Acute toxicity:

Toxic to bees,

Essentially Nontoxic to honeybee

Persistence and Degradability

No data available

Bioaccumulation potential

No data available



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Mobility in Soil

No data available

Results of PBT and vPvB assessment

Assessments not performed

Other adverse effects

None known

13. DISPOSAL CONSIDERATIONS

US EPA WASTE NUMBER & DESCRIPTIONS:

There is no applicable EPA waste number. This product, if discarded, is not expected to be a characteristic or listed hazardous waste.

Disposal Methods

Dispose of product in accordance with federal, state, provincial, and local regulations.

Prevent contamination of environment by wastes.

Pesticide Disposal: To avoid wastes, use all material in this container by application according to label directions. If wastes cannot be avoided, offer remaining product to a waste disposal facility or pesticide disposal program (often such programs are run by state or local governments or by industry).

Container Disposal: Non-refillable container. Do not reuse or refill this container. Triple rinse container (or equivalent) promptly after emptying. Triple rinse as follows: Empty the remaining contents into application equipment or a mix tank and drain for 10 seconds after the flow begins to drip. Fill the container 1/2 full with water and recap. Shake for 10 seconds. Pour rinsate into application equipment or a mix tank or store rinsate for later use or disposal. Drain for 10 seconds after the flow begins to drip. Repeat this procedure two more times. Then offer for recycling if available or puncture and dispose of in a sanitary landfill, or by incineration. Do not burn, unless allowed by state and local ordinances.

14. TRANSPORT INFORMATION

The data provided in this section is for information only. Please apply the appropriate regulations to properly classify your shipment for transportation.

UN No:	Not Applicable
Dangerous Goods Class:	Not Applicable
Packaging Group:	Not Applicable
Marine pollutant:	Not Applicable
ADG Code:	Not a Dangerous Good



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15. REGULATORY INFORMATION

Safety, Health and Environmental Regulations / Specific Legislation

TSCA STATUS:

Exempt

CERCLA STATUS:

Not regulated as hazardous

SARA STATUS:

Acute: No

Chronic: No

Fire: No

CDT: No

RCRA STATUS:

Not regulated as hazardous

State Right-To-Know: Not Listed

Consult applicable national, state provincial or local laws to determine regulations, laws or ordinances which may be applicable.

OSHA HAZARD COMMUNICATION STANDARD:

Not defined by the OSHA Hazard Communication Standard, 29 CFR

EPA Pesticide Regulations

EPA Registration Number: 73049- 46

EPA Pesticide Label signal word: CAUTION

Product must have EPA Approved Pesticide Label attached to or accompanying all containers.

This chemical is a pesticide product registered by the United States Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets (SDS), and for workplace labels of non-pesticide chemicals. The hazard information required on the pesticide label is reproduced below. The pesticide label also includes other important information, including directions and limitations for its use.

KEEP OUT OF REACH OF CHILDREN

CAUTION

PRECAUTIONARY STATEMENTS

HAZARDS TO HUMANS AND DOMESTIC ANIMALS

CAUTION



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Causes moderate eye irritation. Avoid contact with eyes or clothing. Wash thoroughly with soap and water after handling and before eating, drinking chewing gum, using tobacco or using the toilet.

Engineering Controls:

When handlers use closed systems, enclosed cabs, or aircraft in a manner that meets the requirements listed in the Worker Protection Standard (WPS) for agricultural pesticides [40 CFR 170.240(d)(4-6)], the handler PPE requirements may be reduced or modified as specified in the WPS.

IMPORTANT: When reduced PPE is worn because a closed system is being used, handlers must provide all PPE specified above for "applicators and other handlers" and have such PPE immediately available for use in an emergency, such as spill or equipment, breakdown.

Environmental Hazards

For Ground Application:

For Terrestrial Uses: Do not apply directly to water, or to areas where surface water is present or to intertidal areas below the mean high water mark. Do not contaminate water when cleaning equipment or disposing of equipment washwaters.

No manual application can take place within 300 feet of any threatened or endangered Lepidoptera.

For Aerial Application:

Except under the forest canopy, do not apply directly to water, or to areas where surface water is present or to intertidal areas below the mean high water mark. Do not contaminate water when cleaning equipment or disposing of equipment washwaters.

This product must not be applied aurally within ¼ mile of any habitats of threatened or endangered Lepidoptera.

16. OTHER INFORMATION

NFPA Hazard Ratings

Health: 1

Flammability: 0

Instability: 0

HMIS Hazard Ratings

Health: 1

Flammability: 0

Instability: 0

0 = Minimal

1 = Slight

2 = Moderate

3 = Serious

4 = Extreme

REASON FOR ISSUE: New GHS format

APPROVAL DATE: 06/25/15

SUPERSEDES DATE: --/--/--



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LEGEND: N/A = Not Applicable N/D = Not Determined
 N/L = Not Listed L = Listed
 C = Ceiling S = Short-term
(R) = Registered Trademark of THEP WATANA CO., LTD.
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This Safety Data Sheet (SDS) serves different purposes than and DOES NOT REPLACE OR MODIFY THE EPA-APPROVED PRODUCT LABEL (attached to and accompanying the product container). This SDS provides important health, safety, and environmental information for employers, employees, emergency responders and others handling large quantities of the product in activities generally other than product use as required by the Occupational Health and Safety Act (29 CFR 1910.1200, "Hazcom"). The product label provides information specifically for product use in the ordinary course. Use, storage and disposal of pesticide products is regulated by the EPA under the authority of FIFRA through the product label. All necessary hazard classification and appropriate precautionary use, storage, and disposal information is set forth on that label or labeling accompanying the pesticide or to which reference is made on the label. It is a violation of federal law to use an EPA-registered pesticide product in any manner inconsistent with its labeling.